Deep Merge: Deep-Learning-Based Region Merging for Remote Sensing Image Segmentation

Xianwei Lv, Claudio Persello, *Senior Member, IEEE*, Wangbin Li, Xiao Huang, Dongping Ming, *Senior Member, IEEE*, and Alfred Stein

***Abstract*— Image segmentation represents a fundamental step in analyzing very high-spatial-resolution (VHR) remote sensing imagery. Its objective is to partition an image into segments that best match with geo-objects. However, the diverse appearances of geospatial objects often lead to interobject homogeneity and intraobject heterogeneity. Existing segmentation methods often struggle to accurately segment geo-objects with varying shapes and scales. To address these challenges, we propose DeepMerge, a novel method that integrates deep learning and region adjacency graphs (RAGs) to accurately segment complete geo-objects in large VHR images. DeepMerge begins with an initial over-segmentation of the image and then iteratively merges similar regions to achieve complete geo-object segmentation. A deep learning model is employed to learn the similarity between adjacent superpixel pairs. This approach only requires labels indicating whether adjacent superpixels belong to the same geo-object eliminating the need for object-level annotations, enabling weakly supervised segmentation. A cross-scale module is incorporated to capture multiscale information, enhancing the representation of superpixels. In addition, the feature distances between neighboring super-pixels are deemed as scale parameters (thresholds) to control the merging procedure, thus yielding an interpretable, predictable, stable, and optimal scale parameter 0.5. DeepMerge can achieve high segmentation accuracy in a weakly supervised manner, which is validated on large-scale remote sensing images of 0.55-m resolution covering an area of 5660 km^2. The experimental results demonstrate that DeepMerge achieves the highest F value (0.9552) and the lowest total error (TE) (0.0827), accurately segmenting geo-objects of varying sizes and outperforming all competing methods.**

***Index Terms*— Deep learning, image segmentation, region adjacency graph (RAG), region merging, scale parameter interpretability.**

Received 17 January 2024; revised 25 July 2024 and 18 December 2024; accepted 19 February 2025. Date of publication 24 February 2025; date of current version 7 March 2025. This work was supported in part by China Scholarship Council, Scientific Research Start-Up Fund of Northeastern University at Qinhuangdao under Grant 9060212362301; and in part by the Fundamental Research Funds for the Central Universities under Grant 10011236022401. *(Corresponding author: Wangbin Li.)*

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Digital Object Identifier 10.1109/TGRS.2025.3544549

I. Introduction

THE acquisition of very high-spatial-resolution (VHR) remote sensing imagery over vast areas have become increasingly accessible, driven by advancements in satellite remote sensing technology and the adoption of open data policies. Compared to medium and low-resolution images, VHR images can provide richer details and spatial relationships for characterizing geo-objects, and are widely applied in land-cover/land-use classification [1], urban functional zone understanding [2], building roof modeling [3], large-scale terrain classification [4], and object extraction and monitoring [5], [6]. Fundamentally, these applications rely on the accurate identification and semantic interpretation of geo-objects within the imagery.

Deep-learning-based semantic segmentation models can classify pixels into geo-objects within images [7]. However, it requires giant precisely labeled training data, making the labeling process time-consuming and labor-intensive. Furthermore, while commonly applied to land-cover/land-use classification, their application to other remote sensing image interpretation tasks is often limited.

Image segmentation aims to group pixels into meaningful segments representing geo-objects [8], distinguishing it from the pixel-wise classification approach used in semantic segmentation. Consequently, researchers have developed numerous efficient superpixel segmentation methods to facilitate image segmentation. However, the complex spatial relationships and rich spectral information inherent in geo-objects present significant challenges to achieving the high segmentation accuracy demanded by both research and practical applications.

While conventional superpixel segmentation methods efficiently partition images into regions of similar sizes and shapes, they often fail to segment complete geo-objects in images. To address this limitation, numerous region-merging-based methods have been proposed to achieve complete geo-object segmentation [9]. The method merges initial over-segmented super-pixels into geo-objects based on region adjacency graphs (RAGs) [10], [11]. It involves three key steps: 1) generating an initial over-segmentation of the image into superpixels; 2) constructing an RAG to represent the spatial relationships between adjacent superpixels; and 3) iteratively merging the most similar adjacent superpixels based on the edge weights within the RAG. Region-merging methods thus maintain the efficiency of superpixel segmentation while improving overall segmentation accuracy. Importantly, unlike

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semantic segmentation, which operates at the pixel level, region merging approaches focus on object-level characteristics, enabling the preservation and utilization of crucial shape and structural information inherent in geo-objects.

In the third step of standard region-merging-based image segmentation methods, two key factors govern the outcome: 1) the merging criterion, which determines the similarity between adjacent superpixels, and 2) the scale parameter, which defines the stopping condition for the merging process. Merging criteria typically quantify the similarity between adjacent regions by calculating either the feature change resulting from a hypothetical merge or the feature distance between neighboring superpixels. The scale parameter, typically ranging from 0 to positive infinity $[0, +\infty)$, acts as a user-defined threshold that determines the termination condition for the merging procedure. Regions are merged if similarity $\leq$ scale parameter, and the process terminates when similarity $>$ scale parameter. Therefore, the segmentation accuracy of region-merging methods depends critically on both the quality of the superpixel features used for similarity calculation and the appropriate selection of the scale parameter.

1) Conventional textural and morphological features are insufficient to fully represent super-pixels, and subsequently result in wrong merges due to the interobject consistency and intraobject discrepancy. For example, when two adjacent super-pixels exhibit consistent features and are situated within neighboring geo-objects, they are prone to be merged. Conversely, if two neighboring super-pixels manifest disparate features within a single geo-object, they are unlikely to be merged. This discrepancy arises from the absence of cross-scale observations spanning from micro to macro levels in the super-pixel representation.
2) Inappropriate scale parameters lead to over-segmentation errors (using small scale parameters) or under-segmentation errors (using big scale parameters). Existing methods often rely on a trial-and-error approach for scale parameter optimization [12]. However, due to the variety of application scenarios and merging criteria, a single, universally optimal scale parameter is rarely achievable. Moreover, existing automatic scale parameter selection methods are often tailored to specific objects or datasets, limiting their generalizability.

Focus on addressing the above-mentioned issues, we propose a deep-learning-based region merging method dubbed DeepMerge for large-area VHR remote sensing image segmentation. DeepMerge learns the similarity between adjacent super-pixel pairs, leveraging as weakly supervised training data composed of positive pairs, where both super-pixels belong to the same geo-object, and negative pairs, where they belong to different geo-objects. By learning from these pairwise relationships without explicit object-level annotations, the model can effectively infer the similarity and ultimately output complete geo-objects. Fig. 1 illustrates the segmentation results of the proposed method against state-of-the-art (SOTA) region-merging-based segmentation methods.

To comprehend the cross-scale interrelations (inner scale, super-pixel scale, and environmental scale shown in Fig. 2) of super-pixels, we introduced a cross-scale module, which captures spatial information at multiple scales. Critically, "cross-scale" differs from traditional "multiscale" approaches, as it pertains to object-specific scales in contrast to the fixed scales in the multiscale. Furthermore, segment-based features are integrated with the deep features extracted by the cross-scale module, providing a richer representation of superpixels. During training, DeepMerge learns to represent positive superpixel pairs (belonging to the same object) with a feature distance approaching 0, and negative pairs (belonging to different objects) with a distance approaching 1. This behavior leads to an optimal scale parameter of 0.5, as experimentally validated in Section IV-B. The key contributions of our study are the following.

1) This work presents the first integration of RAGs with deep learning for remote sensing image segmentation, enabling a comprehensive characterization of cross-scale superpixel relationships. Experiments demonstrate that our proposed method achieves high segmentation accuracy, segmenting both small and large geo-objects, and outperforming SOTA methods.
2) The optimal scale parameter in DeepMerge is interpretable and predictable, stabilizing at 0.5 in various scenarios, solving the problem of scale parameter selection and releasing the hands of users in selecting optimal scale parameters.
3) The efficiency of training data preparation facilitated by DeepMerge is notable. In contrast to the laborious task of manually annotating five cityscapes images (an open semantic segmentation dataset [15], in which each image sized at 1.024×2.048 pixels) over a span of 7.5 h, the weakly supervised sample annotation method proposed by DeepMerge, as outlined in this article, enables a proficient operator to annotate a training image measuring 12.800×12.800 pixels ($\approx$78 cityscapes images) within the same timeframe.

II. Related Works About Region-Merging-Based Segmentation

Image segmentation plays an essential step in remote sensing image analysis workflows. Numerous methods have been proposed to improve the segmentation quality of remote sensing images, such as the simple linear iterative clustering method [16], mean-shift [17], and multiresolution segmentation (MRS) [18]. Contour-based methods are an important branch of segmentation methods focusing on the extraction of geo-object boundaries [19], [20], [21]. Single-strategy segmentation methods often struggle to achieve high-quality results. Thus, region-merging-based segmentation has received wide attention. In this section, we review related works concerning the rationale of RAG, merging criteria, and optimization of scale parameters.

A. Rationale of RAG

Following the graph structure, RAG is constructed by taking the initial super-pixels as vertices and the connections between

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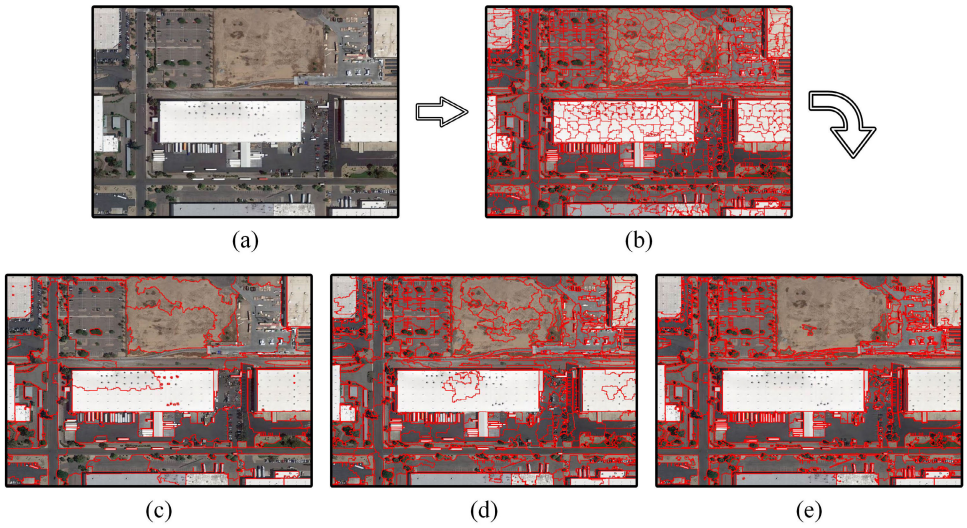

Fig. 1. Comparison of different region-merging-based segmentation methods in region-merging. (a) Original remote sensing image. (b) Initial over-segmented super-pixels. (c) Region-merging results of SOTA method BCMS [13]. (d) Region-merging results of SOTA method Local-SA [14]. (e) Region-merging results of the proposed method DeepMerge.

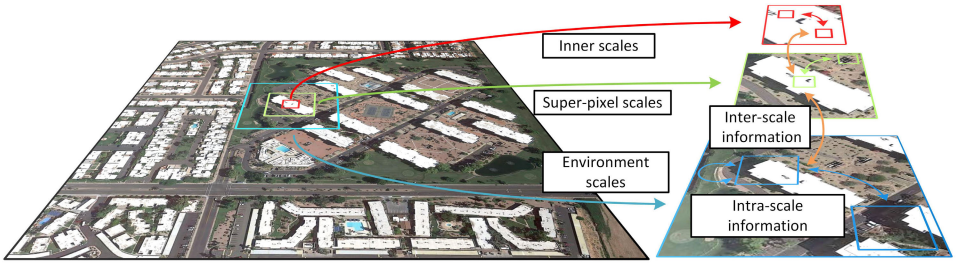

Fig. 2. Cross-scale presentations in the DeepMerge.

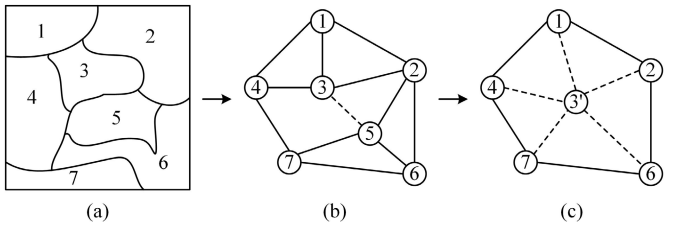

Fig. 3. Construction of RAG and a region-merging step in the RAG. (a) Hypothetical super-pixel segmentations. (b) RAG related to the super-pixels. (c) Hypothetical merging step in the RAG.

super-pixels as edges. Each vertex stores the features of the corresponding super-pixel, and each edge stores the feature distance (i.e., similarity or weight) between two vertices at its ends. Fig. 3 describes the concept of an RAG model in a hypothetical over-segmented case.

Based on the seven over-segmented super-pixels [see Fig. 3(a)], an RAG represented as a nondirectional graph, is constructed. Each vertex in Fig. 3(b) corresponds to a super-pixel from the original segmentation, while edges connecting these vertices represent the spatial adjacency relationships between super-pixels. The weight assigned to each edge quantifies the similarity between two adjacent vertices, determined by a predefined merging criterion. Supposing that the weight value of the dotted edge between v_3 and v_5 in Fig. 3(b) is the smallest, indicating their high similarity, v_3 and v_5 are further merged into a new $v_{3'}$ shown in Fig. 3(c). The weights of edges connected to $v_{3'}$ are then updated. The RAG model iteratively repeats these operations until the smallest weight is higher than a scale parameter.

B. Merging Criteria in RAG

The merging criteria employed in RAG construction exert a significant influence on the quality of segmentation results. Consequently, numerous studies have focused on refining these criteria to improve segmentation accuracy. For instance, Derivaux et al. [22] developed a supervised adaptation of the classic watershed segmentation method by incorporating multispectral gradients. Furthermore, Wassenberg et al. [23] utilized a graph-cutting heuristic method to accelerate the minimum spanning tree-based algorithm. Johnson and Xie [24] introduced an unsupervised segmentation framework that leverages weighted variance and Moran's I, a metric for spatial autocorrelation, across multiple scales to evaluate and refine segmentation outcomes. Other investigations have explored the incorporation of features such as edge strength, compactness, and standard deviation into the merging criteria [9], [13], [25], [26]. Chen et al. [27] devised a multiscale segmentation approach based on stepwise refinement guided by area and boundary features. Wang et al. [28] utilized local spectral statistics to quantify homogeneity and heterogeneity

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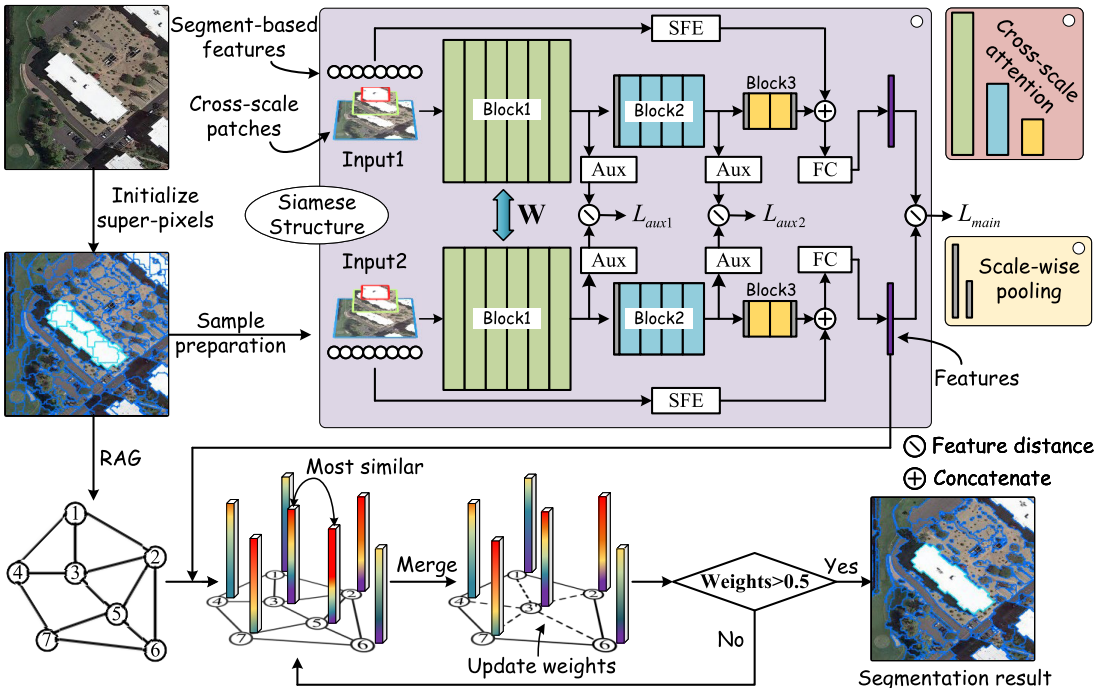

Fig. 4. Workflow of DeepMerge for super-pixel segmentation in high-resolution remote sensing imagery. *W* means shared weights. Aux is the loss function auxiliary module. FC is the fully connected layers. SFE is the segment-based feature embedding module.

between segments. Focusing specifically on the delineation of dune-field landscape patches, Zheng et al. [29] integrated multisource features representing dune-field characteristics at multiple scales. Su et al. [26] treated region merging as a classification task, training a random forest classifier on segment-based features. Despite these advancements, these features often fail to adequately capture the details of geo-objects, potentially resulting in incomplete or inaccurate segmentation of geo-objects.

C. Scale Parameter Optimization in RAG

Considerable efforts have been dedicated to optimizing scale parameters within the RAG. The resulting segmentation output is sensitive to the scale parameter, with variations in this parameter leading to different segmentation outcomes in the investigated area. Low scale results in small-area segments, benefiting the segmentation of small geo-objects; however, large geo-objects tend to contain multiple super-pixels, leading to over-segmentation errors. The manual-optimal scale is often determined by a trial-and-error process, causing uncertainties in segmentation results [1], [30]. Therefore, researchers have increasingly focused on developing automatic and self-adaptive methods for scale optimization. Most region-merging algorithms leverage the homogeneity within and heterogeneity between geo-objects to determine the optimal scale parameter. For example, an automated approach to parameterizing multiscale image segmentation was proposed to detect scale transitions in geo-objects relying on the local variance [31]. Ming et al. [32] introduced a scale parameter selection method for object-based information extraction that utilizes spatial and spectral statistics derived from an average local variance graph. Hu et al. [33] developed a generalized stepwise evolution analysis framework for optimal scale parameter estimation, employing local variance and Moran's I. To accommodate the diverse size range of geo-objects, Zhang et al. [34] introduced a scale-variable segmentation method that adaptively estimates scale parameters. Furthermore, to achieve object-specific optimal scales, Zhang et al. [12] and Shen et al. [35] proposed object-specific optimization strategies that utilize hierarchical tree structures derived from multiscale segmentation.

III. Methodology

A. Outline of the Proposed Method

DeepMerge integrates deep learning and RAG for high-quality segmentation of VHR remote sensing imagery. Fig. 4 summarizes the workflow of DeepMerge. The original image is first over-segmented into super-pixels by a standard segmentation method. Then, we utilize a siamese network [36], [37] to learn the similarity between neighboring super-pixels. We proposed the cross-scale transformer (CSFormer) model as the backbone network. CSFormer is trained using a training set composed of positive and negative samples. Pairs of adjacent super-pixels of the same geo-object are called positive samples. On the contrary, pairs of neighboring super-pixels of different geo-objects are called negative samples. To train the CSFormer, we manually select positive. After training the CSFormer, the model can measure the similarity between adjacent super-pixels. Finally, the RAG model iteratively merges the most-similar super-pixel pairs via the global best-merging strategy until all edge weights exceed 0.5.

For the initial over-segmentation, we recommend MRS, a region-growing algorithm based on the minimum heterogeneity principle [18]. MRS has been proven efficient in

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generating polygonal super-pixels [38]. Of course, other segmentation methods are compatible with our framework for initiating over-segmentation.

We describe the sample collection, cross-scale inputs, segment-based feature embedding, cross-scale attention, and the feature updating strategy for the merging process in Sections III-B–III-D. All codes are written either in C# (sample selection software, cross-scale inputs extraction, and RAG) or Python languages (CSFormer) and tested on a computer with Windows 10 OS, an intel i7-13700K CPU (3.4 GHz), 64-GB RAM, and an NVIDIA GPU (RTX 4090). We open-source the code at https://github.com/lvxianwei/DeepMerge.

B. Sample Collection for Training CSFormer

CSFormer training involves manually collected samples, following these steps: 1) visual analysis of the initial over-segmentation results to identify dominant categories within the study area; 2) selection of adjacent super-pixels with high homogeneity as positive samples [see Fig. 5(b)] and adjacent super-pixels belonging to different categories as negative samples [see Fig. 5(c)], as exemplified in Fig. 5(a). It is worth noting that we need to collect more negative than positive samples to account for the large variety of possible discrepancies between two different super-pixels. In VHR images, one geo-object may contain high heterogeneous super-pixels and different adjacent super-pixels can contain similar pixels, leading to segmentation errors. Therefore, it is necessary to select many samples of both situations. Emphasis is placed on collecting samples from visually similar, yet semantically distinct, adjacent super-pixels (e.g., rivers and vegetation, asphalt roads and shadows, vegetation, and roads). A custom-developed graphical interface facilitates efficient sample collection by automatically recording selected sample pair indices. Using this interface, a skilled operator can collect at least 10 000 sample pairs per day, greatly improving work efficiency.

The CSFormer, like standard deep learning models, requires fixed-size image patches as input. However, the collected super-pixel samples (see Fig. 5) exhibit varying shapes and sizes, creating an input mismatch. To overcome the disparity, we represent super-pixels using multiscale patches capturing both global and local information. We employ a binary tree sampling (BTS) method, previously utilized in object-based convolutional neural networks [30], [39], [40], to generate suitable inputs for the CSFormer. BTS recursively partitions a super-pixel into two subregions until a predefined threshold is reached [40]. This partitioning allows for the extraction of cross-scale data with adaptively sized windows, as illustrated in Fig. 6. For a detailed description of BTS, please refer to [40].

Key parameters in the BTS procedure are the partitioning threshold and the extraction window size. We introduce an automatic strategy to determine these parameters. Following [40], we use up to three patch locations to represent a super-pixel. The window size for each patch is determined by the ratio of the window's intersection area with the super-pixel. Therefore, each super-pixel corresponds to different window sizes. In order to obtain local and global information of super-pixels, we designed extraction windows in three scales (P_1, P_2, and P_3). The automatic strategy is summarized in Algorithm 1.

Algorithm 1 Cross-Scale Input Data Generation

input : A super-pixel *Seg* and a centre position *Pos* for extracting image patches
output: Three patches P_1, P_2, P_3 in three scales

```
patch ← ExtractPatch(Pos, 5);
iter ← 0 ;
while P1, P2== null do
    intersect_area ← Seg.Intersect(patch).area;
    ratio ← intersect_area / patch.area;
    if ratio <= 0.90 and P1 == null then P1 ← patch;
    if ratio <= 0.30 and P2 == null then P2 ← patch;
    iter←iter+1;
    patch ← ExtractPatch(Pos, 5 + iter × 5);
P3 ← ExtractPatch(Pos, P2.width + (P2.width − P1.width)) ;
final;
return P1, P2, P3;
```

Because over-segmentation often results in a single geo-object spanning multiple super-pixels. Thus, information from the P_1 and P_2 may represent the geo-object partially. To enhance the representation of global geo-objects, information from the P_3 needs to be extracted. In certain cases, some small super-pixels are completely segmented without over-segmentation errors. In this situation, the designed cross-scale information strategy provides not only global information for better overall representation but also neighboring information that improves the robustness of region merging. Thus, small and large super-pixels can be well-represented following the designed protocol.

C. CSFormer Model

We propose a Siamese network architecture, with a CSFormer backbone, to learn the similarity between adjacent super-pixels [37]. As illustrated in Fig. 4, the Siamese network utilizes a weight-shared backbone to extract features from positive and negative sample pairs. The features of negative and positive samples can be extracted by a weight-shared backbone. Similarity is determined by a loss function that calculates the distance between feature representations. The CSFormer comprises cross-scale attention modules, scale-wise pooling, auxiliary modules, and segment-based feature embedding modules shown in Fig. 7. These modules are detailed in the following sections.

1) Cross-Scale Attention Mechanism: Cross-scale attention, the core of the CSFormer block, captures long-range and cross-scale pixel dependencies based on the self-attention mechanism with 3-D relative position embedding (see Fig. 8). Inspired by the human visual system's ability to prioritize

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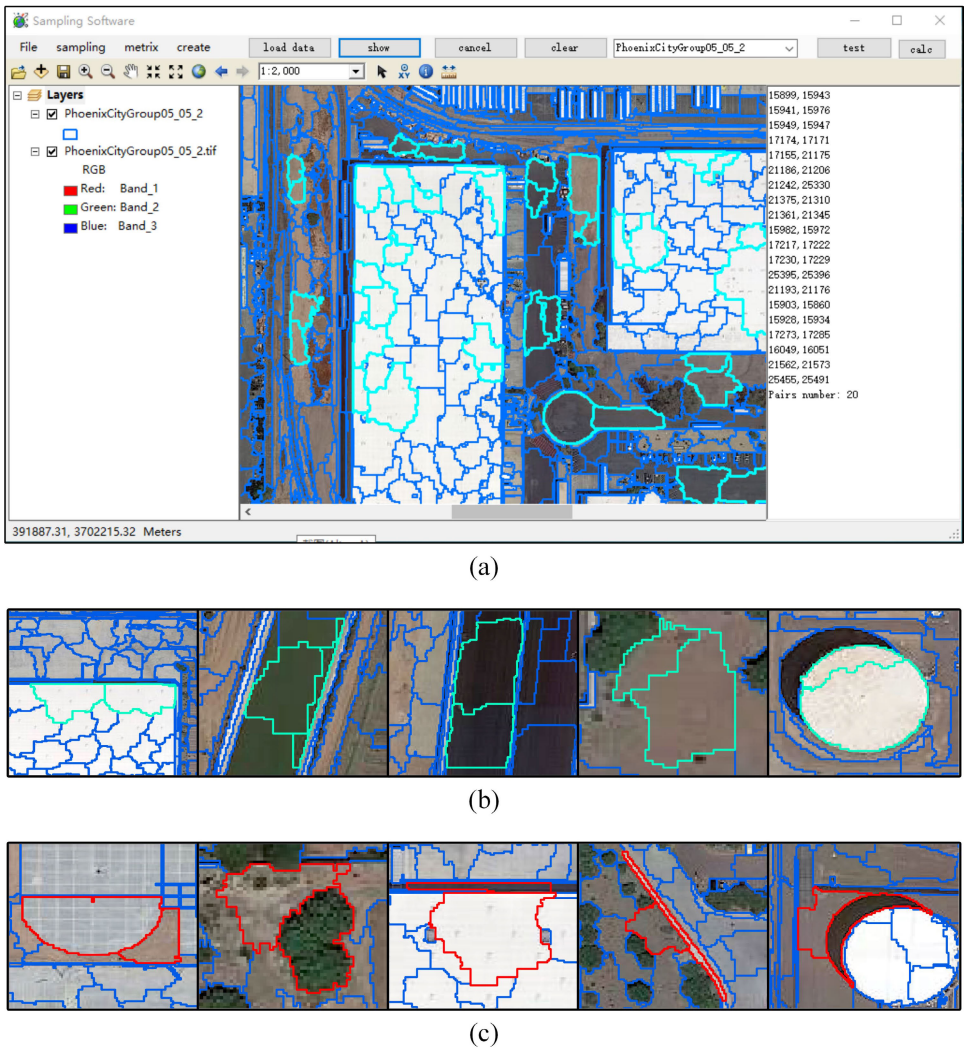

Fig. 5. Sample collection graphical interface and sample pairs. (a) Graphical interface of sample collection. (b) and (c) Positive sample pairs and negative sample pairs, respectively.

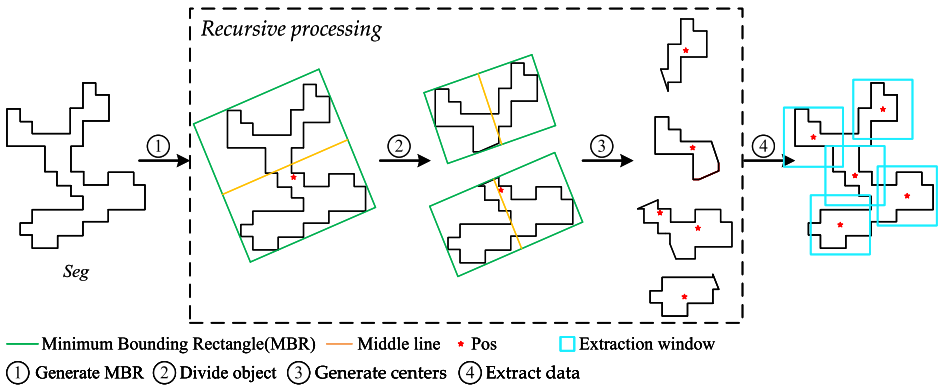

Fig. 6. Basic theory about BTS.

salient information [41], this mechanism facilitates efficient and accurate feature extraction. The CSFormer is a feature extraction structure composed of multihead modules using a cross-scale attention mechanism. The input patches first are resized into 32×32, 64×64, and 128×128, then embedded into three 8×8 vectors of features via cross-scale embedding module 2-D convolution functions (Conv2D), whose kernel sizes are 4×4, 8×8, and 16×16, respectively. These embedded features are flattened, concatenated, and fed into the cross-scale attention modules. The linear is a one-layer fully connected network. MLP denotes the multilayer perception, and the Gaussian error linear units (GELUs) is the activation

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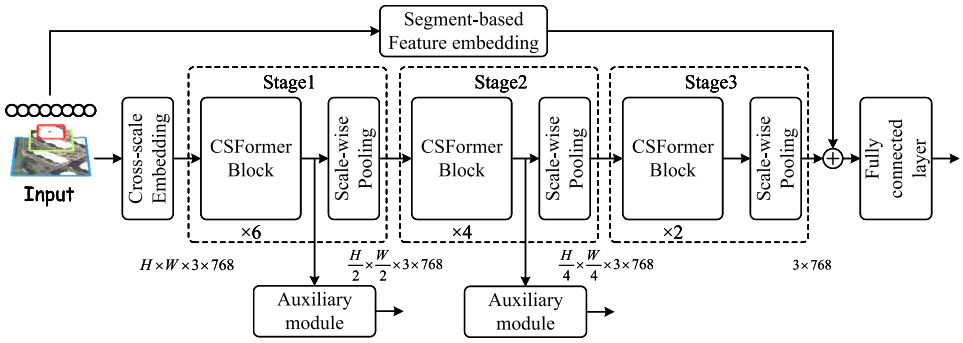

Fig. 7. Architecture of the CSFormer in the DeepMerge.

function [42]. The layer Dropout is employed as a regularization strategy.

Equation (1) depicts the process of the attention mechanism. $V(=\mathrm{AW}^v)$ is the output of the input $A\{a_1, a_2, a_3\}$ (see Fig. 8) linear transformed by the weight W^v. In visual image processing, A is a 1-D vector composed of embedded features a_1, a_2, and a_3. The softmax function plays the role of scorer for V based on query ($Q = \mathrm{AW}^q$) and key ($K = \mathrm{AW}^k$), where W^q and W^k are the weights for updating in the backpropagating process, Q and K are two special matrices used for searching the importance of pixels. Inspired by the 2-D relative-position embedding, we expand it to a 3-D relative-position embedding to measure the relative spatial relationships B of the cross-scale inputs. The features of the self-attention output are finally fed into an MLP

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^T}{\sqrt{d_k}} + B\right)V \tag{1}$$

$$\mathrm{softmax}(z_i) = \frac{\exp(z_i)}{\sum_{j=1}^{K}\exp(z_j)} \tag{2}$$

where d_k is the dimension of K. The output of the attention module, named head, contains scored information in A. The above process is the basic principle of the cross-scale attention mechanism. The softmax function applies the standard exponential function to each element z_i of the input vector z and normalizes these values by dividing by the sum of all these exponentials; this normalization ensures that the sum of the components of the output vector z is 1. The cross-scale attention forms the CSFormer block in Fig. 7.

2) Scale-Wise Pooling: To reduce model complexity, we introduce scale-wise pooling, which pools the CSFormer block outputs at three scales. As depicted in Fig. 9, the CSFormer block output is a 1-D feature vector comprising three scales (represented by red, green, and blue). These features are first reshaped into their respective 2-D representations. The average pooling is then applied to reduce the size of 2-D feature maps. Finally, the resulting 2-D features are flattened and concatenated into a 1-D vector, effectively halving its size.

3) Auxiliary Module: The auxiliary module composed of convolutional blocks (Conv), batch normalization (BN), the linear rectification function (ReLU), and a global pooling layer (AvgPool), can assist in improving the performance and robustness of the CSFormer. The 1-D features of the CSFormer block are processed individually in three branches related to the three scales shown in Fig. 10.

4) Segment-Based Feature Embedding: A segment-based feature embedding block encodes segment-based features, complementing the deep features extracted by networks composed of a Conv1d-GeLU-Conv1d-Conv1d-BN layer. Eighteen segment-based features, including texture, statistical, and shape descriptors, as well as spectral statistics (mean and standard deviation of each band), compactness, brightness, border strength, and resizing factors, are calculated for each segment. These features are computed considering the pixels within the segment. They capture different characteristics to assess region similarities with respect to deep learning features. Their calculations follow:

$$\mathrm{Mean}_i = \frac{1}{n}\sum_{j=1}^{n} v_j \tag{3}$$

$$\mathrm{Std}_i = \sqrt{\frac{1}{n-1}\sum\nolimits_{j=1}^{n}\left(v_j - \mathrm{Mean}_i\right)^2} \tag{4}$$

$$\mathrm{Shape} = \frac{l}{4\sqrt{C}} \tag{5}$$

$$\mathrm{Compactness} = l\sqrt{n} \tag{6}$$

$$\mathrm{Brightness} = \frac{1}{w^B}\sum_{i=1}^{K} w_i^B \mathrm{Mean}_i \tag{7}$$

$$\mathrm{Border} = \frac{l}{2(\mathrm{length} + \mathrm{width})} \tag{8}$$

where i ($i = 1, 2, 3$) indicates the ith band, n denotes the super-pixel size in pixels, v_j denotes the jth pixel value, and Mean_i denotes the mean value of the ith band in a super-pixel. Std_i denotes the standard deviation of the ith band in a super-pixel. Shape denotes the shape indicator defined by the perimeter (l) of a super-pixel and the perimeter (C) of the minimum bounding rectangle (MBR) of a super-pixel. Length and width are the long edge and the short edge of the MBR. Length and width are used to define the border indicator (border) and compactness. K denotes the number of bands, and w_i^B denotes the related weight.

5) Loss Function: The output of the CSFormer is a 1-D vector storing features used for super-pixel representation, and it will be used in calculating similarity in an RAG model. The features of positive sample pairs or negative sample pairs are extracted by the network and are supervised by the loss

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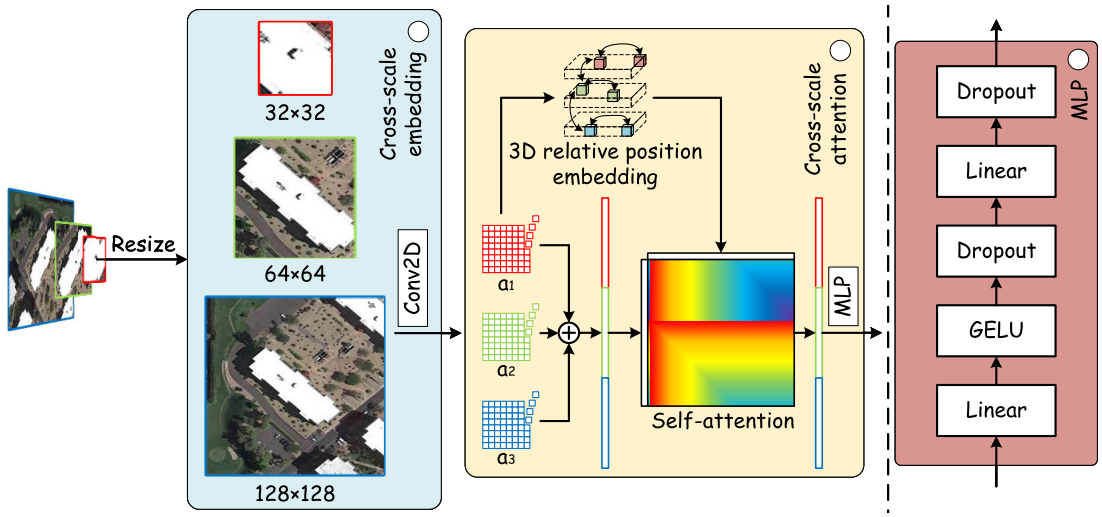

Fig. 8. Cross-scale attention mechanism.

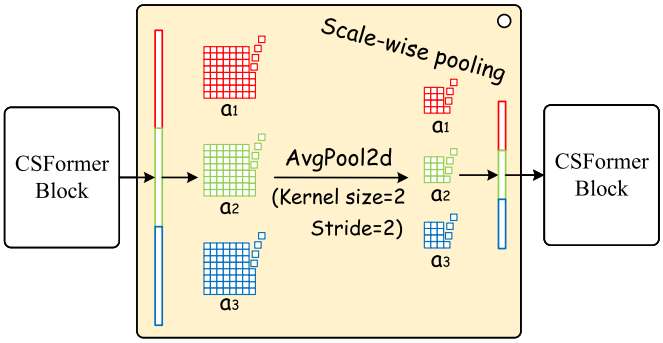

Fig. 9. Scale-wise pooling module.

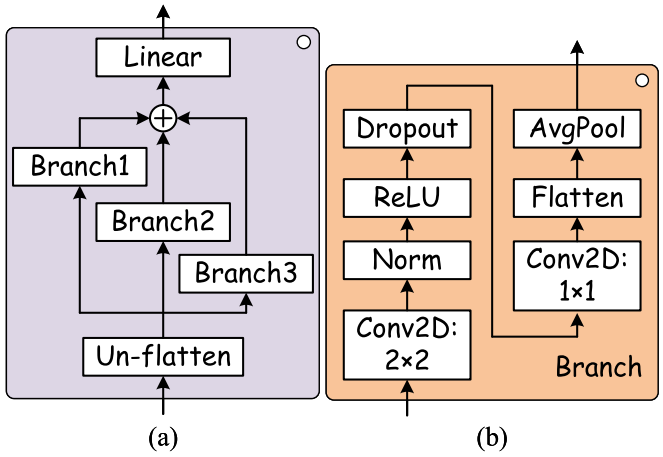

Fig. 10. (a) Auxiliary module and (b) branch of the auxiliary module.

function. Given a pair of super-pixels, the L_{aux1}, L_{aux2}, and L_{main} in Fig. 4 are calculated by the loss function

$$\text{Loss} = \alpha \left\| A_i^{\text{left}} - A_i^{\text{right}} \right\|_2 + (1-\alpha) \max\left(0, \left(\lambda - \left\| A_i^{\text{left}} - A_i^{\text{right}} \right\|_2\right)\right) \tag{9}$$

where A means the feature vector in a super-pixel, i indicates the feature item from the A. α is a binary indicator (positive pair: $\alpha = 1$; negative pair: $\alpha = 0$). λ is a user-defined parameter that represents the cluster center of the Euclidean distance in feature space between negative sample pairs (we recommend $\lambda = 1$). The function max(para1, para2) can return the maximum value from two parameters. In the proposed DeepMerge, the distance in feature space between positive sample pairs should be close to 0, while the distance between negative sample pairs should be close to 1 if less than 1, which predicts that the optimal scale parameter of the proposed method will be 0.5. Once the super-pixel features are extracted by the CSFormer, they further serve as the vertex features in the RAG model (described in Fig. 4). The final loss of the whole CSFormer is calculated by the sum of the weighted loss values above as follows:

$$L_{\text{final}} = L_{\text{main}} + \lambda_1 \times L_{\text{aux1}} + \lambda_2 \times L_{\text{aux2}} \tag{10}$$

where λ_1 and λ_2 indicate the weights of L_{aux1} and L_{aux2}.

D. Merging Criteria and Feature Updating

Merging criteria are designed for calculating the similarity between super-pixels. We use the Euclidean distance of features from two neighboring super-pixels as the merging criteria in the proposed method

$$\text{MC} = \left\| A_i^{\text{left}} - A_i^{\text{right}} \right\|_2 \tag{11}$$

where MC is the Euclidean distance of two group features, which is the similarity of two super-pixels, recorded as the weight of the connected edge in the RAG model. Conventional merging criteria update edge weights of a newly merged super-pixel by reextracting features and recalculating weights between the super-pixel and its adjacent super-pixels. However, deep features of a newly merged super-pixel can cause inefficiency in region-merging if they are reextracted by the pretrained model. In the DeepMerge, the features of a new super-pixel are calculated via the weighted average of the original two features by the following equation:

$$A_i^{\text{left+right}} = \frac{1}{m+n}\left(m A_i^{left} + n A_i^{\text{right}}\right) \tag{12}$$

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TABLE I
ASSESSMENT METRICS USED FOR SEGMENTATION ACCURACY ESTIMATION

Assessment metrics	Formulas	Range	Trend
precision	$\|S\| = \sum_{i=1}^{M} \|S_i\|$ $precision = \sum_{i=1}^{M} \|S_i \bigcap R_{i,\max}\| / \|S\|$	[0,1]	↑
recall	$\|R\| = \sum_{i=1}^{N} \|R_i\|$ $\|R\| = \sum_{i=1}^{N} \|R_i\|$ $recall = \sum_{i=1}^{N} \|R_i \bigcap S_{i,\max}\| / \|R\|$	[0,1]	↑
F	$F = 1/\left(\beta \frac{1}{p} + (1-\beta)\frac{1}{r}\right)$	[0,1]	↑
$GOSE$	$GOSE = \frac{1}{\|R\|} \sum_{i=1}^{N} \|R_i\| \frac{\|R_i \setminus S_{i,\max}\|}{\|R_i\|-1}$	[0,1]	↓
$GUSE$	$GUSE = \frac{1}{\|R\|} \sum_{i=1}^{N} \min\left(\|R_i \bigcup S_{ij}\| \setminus \|R_i \bigcap S_{ij}\|, \|R_i\|\right)$	[0,1]	↓
TE	$TE = GOSE + GUSE$	[0,2]	↓

where $A_i^{\text{left+right}}$ are the features of new super-pixels. m and n are the number of extracting centers by BTS from the super-pixel left and right, denoting corresponding feature vector weights. Therefore, the feature vector weight of the newly merged super-pixel is $m + n$. In fact, calculating the features of the newly merged super-pixel is calculating the spatial clustering center of the features of all initial super-pixels it contains. The feature vector of an initial super-pixel is the average of features extracted from multiple extracting centers by BTS.

E. Segmentation Accuracy Estimation

To assess the segmentation performance of the proposed DeepMerge, two sets of accuracy assessment metrics were employed, focusing on over-segmentation, under-segmentation, and overall segmentation. The first set, as described by [43] comprises precision, recall, and F value. The second set of metrics [44] includes the global over-segmentation error (GOSE), global under-segmentation error (GUSE), and total error (TE). Higher precision and lower GUSE values indicate stronger over-segmentation errors and weaker under-segmentation errors. These metrics, requiring polygonal segmentation outputs and reference geo-objects, have demonstrated effectiveness and robustness in evaluating local and global segmentation performance from multiple perspectives. The calculations of these metrics are presented in Table I. S is the set of polygonal segmentation results containing M segments $\{S_1, S_2, \ldots, S_M\}$, and R is the set of polygonal reference geo-objects containing N reference geo-objects $\{R_1, R_2, \ldots, R_N\}$. $|*|$ represents the area of a segment. $R_{i,\max}$ denotes the largest area reference geo-object related to the ith segment R_i, and $R_{i,\max}$ denotes the largest area segment related to the reference geo-object R_i. R_{ij} denotes the set of segments related to R_i. $S_i \cap R_{i,\max}$ is the intersection of S_i and $R_{i,\max}$, and the $S_i \cup R_{i,\max}$ the union of them. The difference set $|R_i \setminus S_{i,\max}|$ contains pixels in R_i, but not in $S_{i,\max}$. β is set as 0.5. The ↑ means higher values with better performance and vice versa for the ↓.

IV. EXPERIMENTAL RESULTS

A. Dataset

The study area encompasses nine cities within the Phoenix metropolitan area, AZ, USA, namely, Phoenix, Glendale, Scottsdale, Tempe, Mesa, Chandler, Peoria, Surprise, and Goodyear. The images are composites from Google Earth [45] with a variety of sensors (e.g., WorldView, QuickBird, and IKONOS) captured at different times. The image contains 18.7B pixels (182.272 × 102.626, covering 5660 km^2) with 0.55-m resolution and RGB bands encoded as 8-bit integral values. The imagery includes a variety of land cover types, such as urban residential areas, urban green spaces, industrial zones, rural farmlands, water bodies, and bare lands (see Fig. 11).

We partition the image into a total of 135 tiles encoded by row and column numbers in Fig. 11. The average size of a tile is 167M (12.800 × 12.800) pixels. In the Phoenix dataset, negative and positive pairs are selected in the patches with blue masks, as shown in Fig. 11. A total of 71 948 super-pixel pairs (54 945 negative sample pairs and 17 003 positive sample pairs) involving 92 421 super-pixels are manually collected as training data. There is a total of 47 995 928 super-pixels in the whole study area. The training data accounts for 0.19% (≈92 421/47 995 928) of the total number of super-pixels.

To assess the performance of DeepMerge, a total of 3776 polygons are manually digitized as reference geo-objects. To validate the transferability of DeepMerge, the reference geo-objects, covering a variety of land uses (see Fig. 12), are selected in the green mask tiles in Fig. 11.

To ensure consistency, the parameters shape, compactness, and scale for the MRS algorithm, utilized for initial

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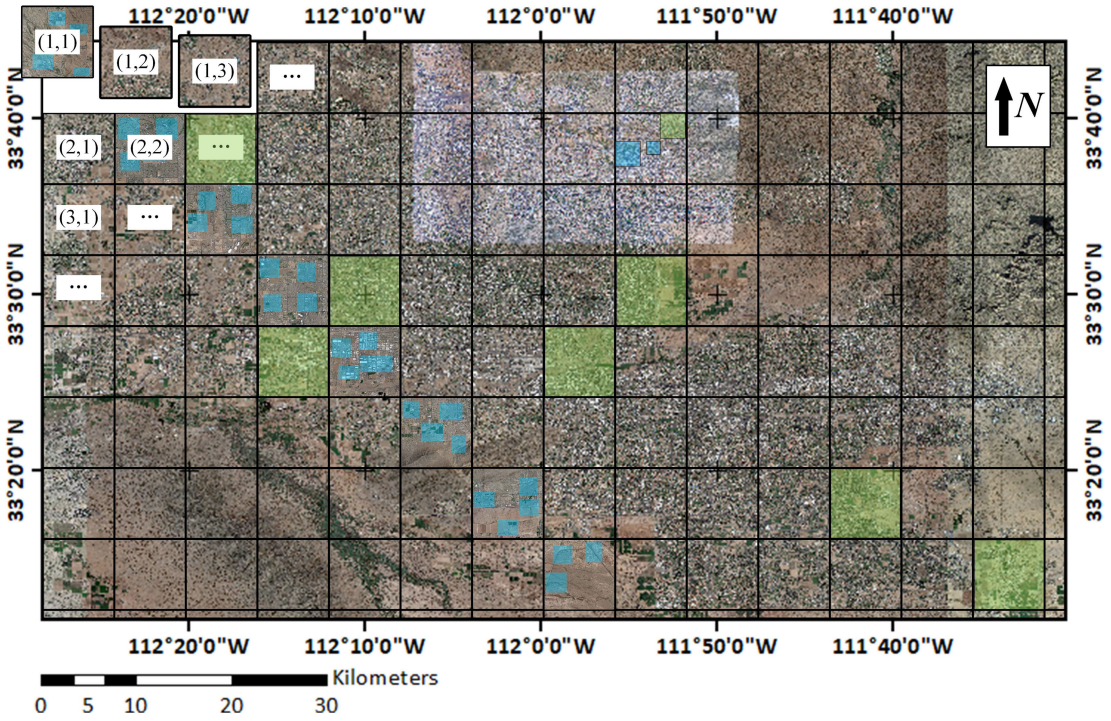

Fig. 11. Study area of the Phoneix city cluster. The translucent blue tiles are used for training and the green tiles are used for accuracy assessment.

Fig. 12. Examples of reference geo-objects.

over-segmentation, are set to 0.5, 0.5, and 25, respectively. An excessive number of segments can hinder the efficiency of region-merging. The MRS, with these parameters, effectively segmented the minimum geo-objects in the dataset with minimal over-segmentation errors, thus balancing the tradeoff between over-segmentation reduction and superpixel quantity.

B. Validation for the Optimal Scale Parameter Predicted in DeepMerge

Sections III-C5 and III-D establish the scale parameter as the feature distance between sample pairs. During CSFormer training, the feature distances between positive and negative pairs progressively converge toward 0 and 1, respectively. Consequently, the feature clustering process suggests an optimal scale parameter of 0.5 for the proposed method, representing the midpoint between 0 and 1. To validate this predicted optimal scale parameter, a sensitivity analysis was conducted using a range of scale parameters: 0.01, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, and 0.99. The segmentation evaluation metrics of precision, recall, and F value of the proposed DeepMerge are presented in Fig. 13(a). The curve of precision [red line in Fig. 13(a)] tends to decrease slowly for scale parameters from 0.0 to 0.6. After 0.6, however, the precision decreases sharply with the further increase of scale parameters. The recall (green line) of the proposed DeepMerge increases sharply when the scale parameter is above 0.2. In comparison, the global F value (blue line) of the whole study area first increases and then decreases with the continuous increase of scale parameters. The highest F value is achieved when the scale parameter is 0.5. Fig. 13(b) depicts the global F value and local F values in green mask tiles (see Fig. 11) in the varying scale parameters, where the gray curves of local F values first tend to increase and then decrease like the global F value. All of the optimal scale parameters of local F values are distributed around 0.5. These findings confirm that the optimal scale parameter for DeepMerge is indeed 0.5.

C. Ablation Experiments

The DeepMerge parameters, including margin value, training epochs, batch size, and learning rate, are set to 1.0, 100, 120, and 0.0001, respectively. Table II presents the quantitative segmentation performance of the ablation experiments using six evaluation metrics. The abbreviation "CSE" denotes a model incorporating a cross-scale embedding module, while "3DP" indicates the inclusion of a 3-D position embedding module. As anticipated, the "CSE" model achieved commendable performance, effectively learning cross-scale geo-object features. "Aux" denotes a model with an auxiliary module added to the backbone. "SEF" denotes a model with a segment-based feature module added to the backbone. The "depth" depicts the layer number of stages in CSFormer. The results suggest desirable performance, evidenced by the high recall and F values and low GOSE, TE values, proving

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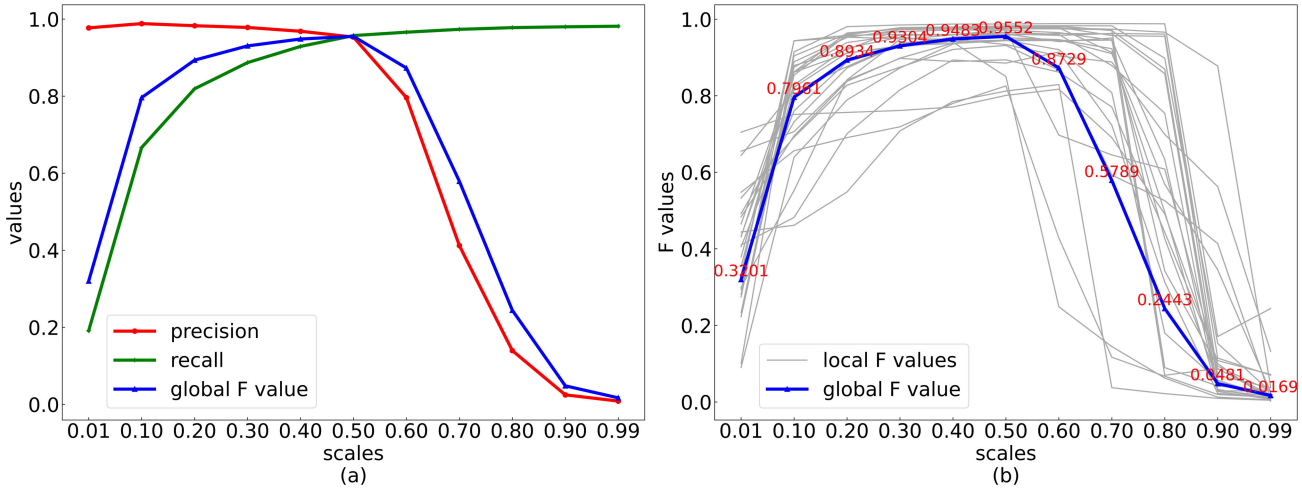

Fig. 13. Segmentation performance of the proposed DeepMerge with different scale settings. (a) Performance curves by precision, recall, and F value. (b) Performance curves by global F values in the whole study area and local F values in the green mask study areas.

the ability of CSFormer to improve the image segmentation performance.

The optimization of weights λ_1 and λ_2 in (10) are shown in Table III. We choose a series of values 0.05, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, and 0.90 as weight candidates for λ_1 and λ_2. F value is used as an indicator for measuring the performance of the combination of λ_1 and λ_2 in various values. We observe that the highest F value is achieved on the condition of λ_1 and λ_2 equal to 0.3 and 0.4.

D. Image Segmentation Results

Relying on the same initial over-segmentation, ten region-merging-based supervised and unsupervised methods are selected as competing algorithms, including BCMS [13], FHS [9], HRM [34], CSVD [46], USIH [47], Local-SA [14], OMS [35], OSO (fine and coarse) [12], MLRM [26], and IOseg [1]. These methods employ diverse optimization strategies, encompassing merging criteria (e.g., edge penalty), geo-object optimization, and scale sets. The varying merging criteria necessitate different scale parameters. The scale parameters of BCMS and FHS are optimized by global score [24]. The scale parameters in HRM, CSVD, Local-SA, MLRM, and IOseg are set as recommended. USIH, OMS, and OSO automatically generate specific geo-objects with optimized scales, where OSO can produce coarse and fine segmentations at the same time. In our experiments, we set the scale parameter of DeepMerge to 0.5. Beyond traditional image segmentation methods, deep-learning-based semantic segmentation has witnessed substantial advancements in recent years. Decoder–encoder neural networks are able to predict pixel-wise labels in the image. Novel modules have been designed to be applied in various applications to improve semantic segmentation accuracy. Standard deep-learning-based semantic segmentation methods, UNet [48], UNet++ [49], U2Net [50], UNetFormer [51], FCN [52], SegNet [53], DeepLab [54], PspNet [55], ABCNet [56], MAResUNet [57], EANet [58], CCnet [59], SegFormer [60], DenseASPP [61], and ENet [62] are selected as the competing algorithms. We labeled the blue mask areas (see Fig. 11) as the training dataset for semantic segmentation networks. We preserved the boundaries of semantic segmentation resulting in the green mask areas serving as polygonal geo-objects in comparison to DeepMerge. Further details regarding the labeling of the semantic segmentation dataset, network training procedures, and the generation of image segmentation outputs are provided in the Appendix.

The segmentation performance of DeepMerge and the comparative methods is presented in Table IV, with a sensitivity analysis of the scale parameter provided in Section IV-B. The initial segmentation approach, MRS, yielded the highest precision (0.9772) and the lowest GUSE value (0.0346) among all methods evaluated. We notice that the proposed DeepMerge achieves the best performance in the rest metrics compared with other methods. The F value of DeepMerge is 0.9552 higher than IOseg, the second-best method (F value: 0.7981). The TE values (0.0827) of the DeepMerge are the lowest, suggesting small segmentation errors. Note that the proposed DeepMerge, among all competing algorithms, achieves the highest recall value and the lowest GOSE value at the same time, indicating its superiority and robustness. In contrast, the semantic segmentation methods exhibited comparatively weaker performance than the standard superpixel-based methods. The precision and GOSE values of these methods are extremely low, as similar nearby geo-objects can be clustered into the same geo-object, resulting in substantial undersegmentation errors. The F values and TE vales of segmentation results based on the semantic segmentation methods are low and high, respectively, caused by low precision and high GUSE, meaning serious under-segmentation errors in the results caused by the pixel adhesion, whose details are described in the Appendix.

Representative segmentation results of the investigated algorithms are presented in Fig. 14, where three prevalent landscape types are highlighted: 1) urban residential areas [see Fig. 14(a)]; 2) rural industrial zones [see Fig. 14(b)]; and 3) rural green spaces [see Fig. 14(c)]. Initial observations reveal that the results from MRS exhibit noticeable

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TABLE II
ABLATION EXPERIMENTS ON MODEL VARIATIONS

CSE	3DP	Aux	SEF	depth	precision↑	recall↑	F↑	GOSE↓	GUSE↓	TE↓
✓				[6,4,2]	0.9142	0.9122	0.9132	0.0534	0.0539	0.1073
✓	✓			[6,4,2]	0.9178	0.9387	0.9281	0.0452	0.0575	0.1027
✓		✓		[6,4,2]	0.9100	0.9408	0.9251	0.0443	0.0574	0.1017
✓			✓	[6,4,2]	0.9320	0.9220	0.9275	0.0513	0.0512	0.1025
✓		✓	✓	[6,4,2]	0.9319	0.9564	0.9440	0.0334	0.0526	0.0860
✓	✓		✓	[6,4,2]	0.9350	0.9485	0.9417	0.0415	**0.0491**	0.0906
✓	✓	✓		[6,4,2]	0.9322	0.9419	0.9370	0.0420	0.0574	0.0994
✓	✓	✓	✓	[6,4,2]	**0.9533**	**0.9572**	**0.9552**	**0.0330**	0.0497	**0.0827**
✓	✓	✓	✓	[3,2,1]	0.9295	0.9463	0.9378	0.0393	0.0635	0.1028

TABLE III
WEIGHT OPTIMIZATION OF THE AUXILIARY MODULE

F value (λ_1 \ λ_2)	0.05	0.10	0.20	0.30	0.40	0.50	0.60	0.70	0.80	0.90
0.05	0.9401	0.9014	0.9279	0.9176	0.9305	0.9317	0.8982	0.9425	0.9344	0.9190
0.10	0.9390	0.9423	0.9550	0.9416	0.9359	0.9178	0.9187	0.9201	0.9371	0.9535
0.20	0.9369	0.9255	0.8926	0.9374	0.9391	0.9146	0.9290	0.9342	0.9230	0.8869
0.30	0.8890	0.9291	0.8956	0.9507	**0.9552**	0.9375	0.9278	0.9400	0.8957	0.8224
0.40	0.9115	0.9313	0.9063	0.9270	0.9114	0.8939	0.9147	0.9415	0.9260	0.9248
0.50	0.9073	0.9224	0.9440	0.8776	0.9350	0.9181	0.9188	0.9396	0.9191	0.9393
0.60	0.9313	0.9232	0.9205	0.8947	0.9320	0.9313	0.9357	0.9436	0.9358	0.9062
0.70	0.9446	0.9263	0.9310	0.8872	0.9235	0.9333	0.9401	0.9319	0.9038	0.9048
0.80	0.9070	0.9048	0.8854	0.9133	0.9430	0.9342	0.8839	0.9333	0.9067	0.9339
0.90	0.9405	0.8883	0.9153	0.9344	0.9385	0.9378	0.9246	0.9232	0.9193	0.9475

TABLE IV
IMAGE SEGMENTATION PERFORMANCE OF THE DEEPMERGE AND OTHER COMPETING METHODS

Method	precision↑	recall↑	F↑	GOSE↓	GUSE↓	TE↓
MRS[18]	**0.9772**	0.1913	0.3200	0.1028	**0.0346**	0.1374
BCMS[13]	0.9620	0.2468	0.3929	0.1127	0.0442	0.1570
FHS[9]	0.5335	0.5115	0.5222	0.1465	0.2517	0.3982
HRM[34]	0.9714	0.3288	0.4913	0.1246	0.0394	0.1641
CSVD[46]	0.9657	0.5954	0.7366	0.1762	0.0504	0.2265
USIH[47]	0.8870	0.6504	0.7505	0.1164	0.0483	0.1647
Local-SA[14]	0.7062	0.8179	0.7580	0.0610	0.1370	0.1980
OMS[35]	0.9733	0.3154	0.4764	0.1145	0.0819	0.1964
OSO(fine)[12]	0.9700	0.4624	0.6262	0.1387	0.0406	0.1793
OSO(course)[12]	0.9586	0.5150	0.6700	0.1283	0.0469	0.1751
MLRM[26]	0.9699	0.3860	0.5522	0.1606	0.0390	0.1996
IOseg[1]	0.7346	0.8737	0.7981	0.0903	0.0563	0.1466
UNet[48]	0.2424	0.9165	0.3834	0.0558	0.4076	0.4634
FCN[52]	0.0364	0.9029	0.0700	0.0615	0.4468	0.5083
SegNet[53]	0.0678	0.9222	0.1263	0.0470	0.4295	0.4765
PspNet[55]	0.0182	0.8810	0.0357	0.0775	0.6564	0.7339
DeepLab[54]	0.0169	0.8751	0.0331	0.0742	0.5394	0.6136
U2Net[50]	0.0677	0.7685	0.1244	0.1289	0.3774	0.5063
EANet[58]	0.0024	0.8264	0.0048	0.1061	0.5206	0.6267
ABCBet[56]	0.0350	0.8032	0.0671	0.1187	0.4549	0.5736
MAResUNet[57]	0.0288	0.7858	0.0556	0.1271	0.3302	0.4573
UNetFormer[51]	0.0019	0.7837	0.0038	0.1324	0.7611	0.8935
CCNet[59]	0.0060	0.8581	0.0120	0.0834	0.5953	0.6787
SegFormer[60]	0.0235	0.8200	0.0458	0.1097	0.4691	0.5788
UNet++[49]	0.1307	0.7692	0.2234	0.1295	0.1718	0.3013
DenseAspp[61]	0.0058	0.6871	0.0114	0.1689	0.5051	0.6740
ENet[62]	0.0302	0.9278	0.0585	0.0458	0.5962	0.6420
Ours (DeepMerge)	0.9533	**0.9572**	**0.9552**	**0.0330**	0.0497	**0.0827**

over-segmentation errors. Leveraging the MRS results as a foundation, the other investigated methods generated their respective optimized segmentations. Notably, the segmentation results produced by DeepMerge are satisfactory and outperform those of the other methods. Almost all house outlines, road boundaries, green spaces, and even sidewalk boundaries are precisely delineated by DeepMerge [see Fig. 14(a)]. FHS and USIH also achieve good performances, however, with discontinuous roads. The superior performance of DeepMerge is further exemplified in Fig. 14(b), where geo-objects exhibit

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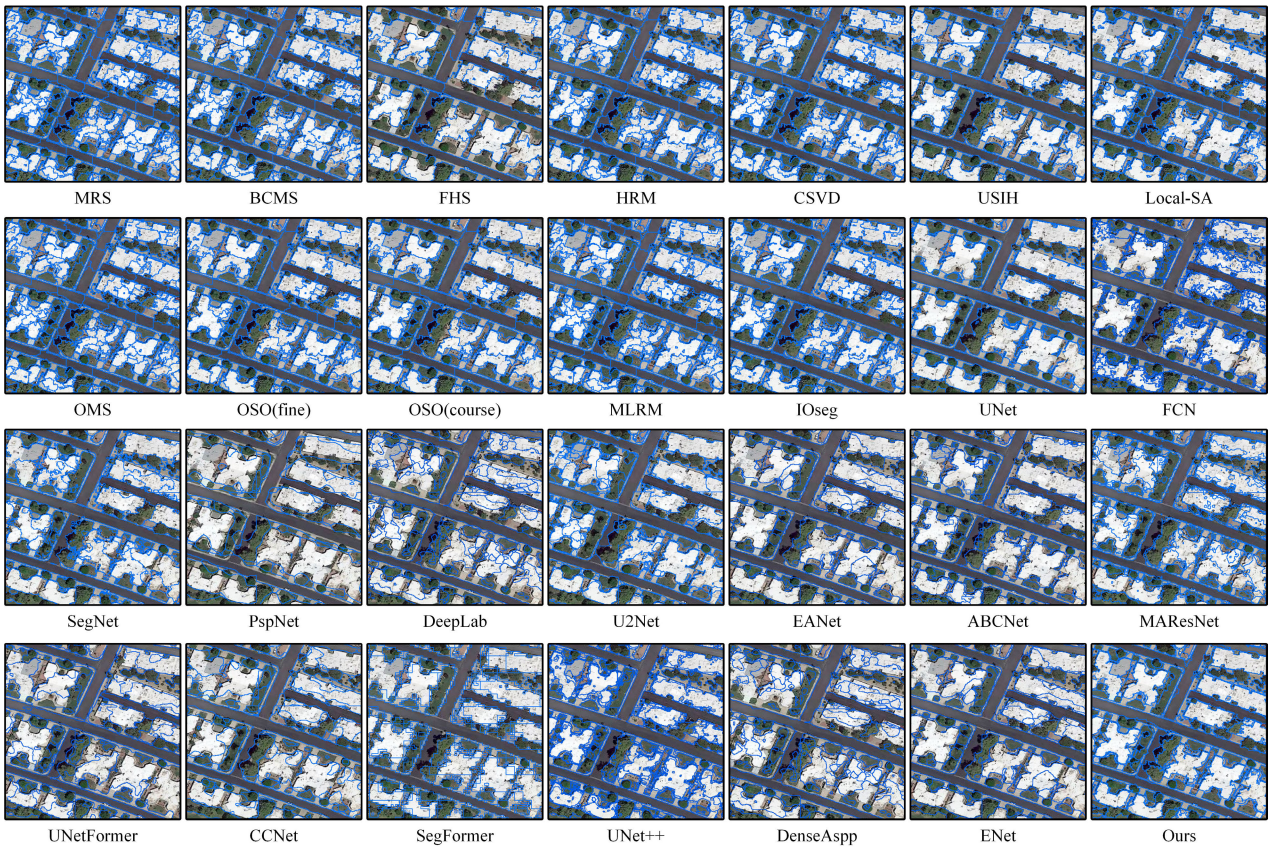

(a)

(b)

Fig. 14. (a) Image segmentation results in urban residential areas. (b) Image segmentation results in rural industrial zones.

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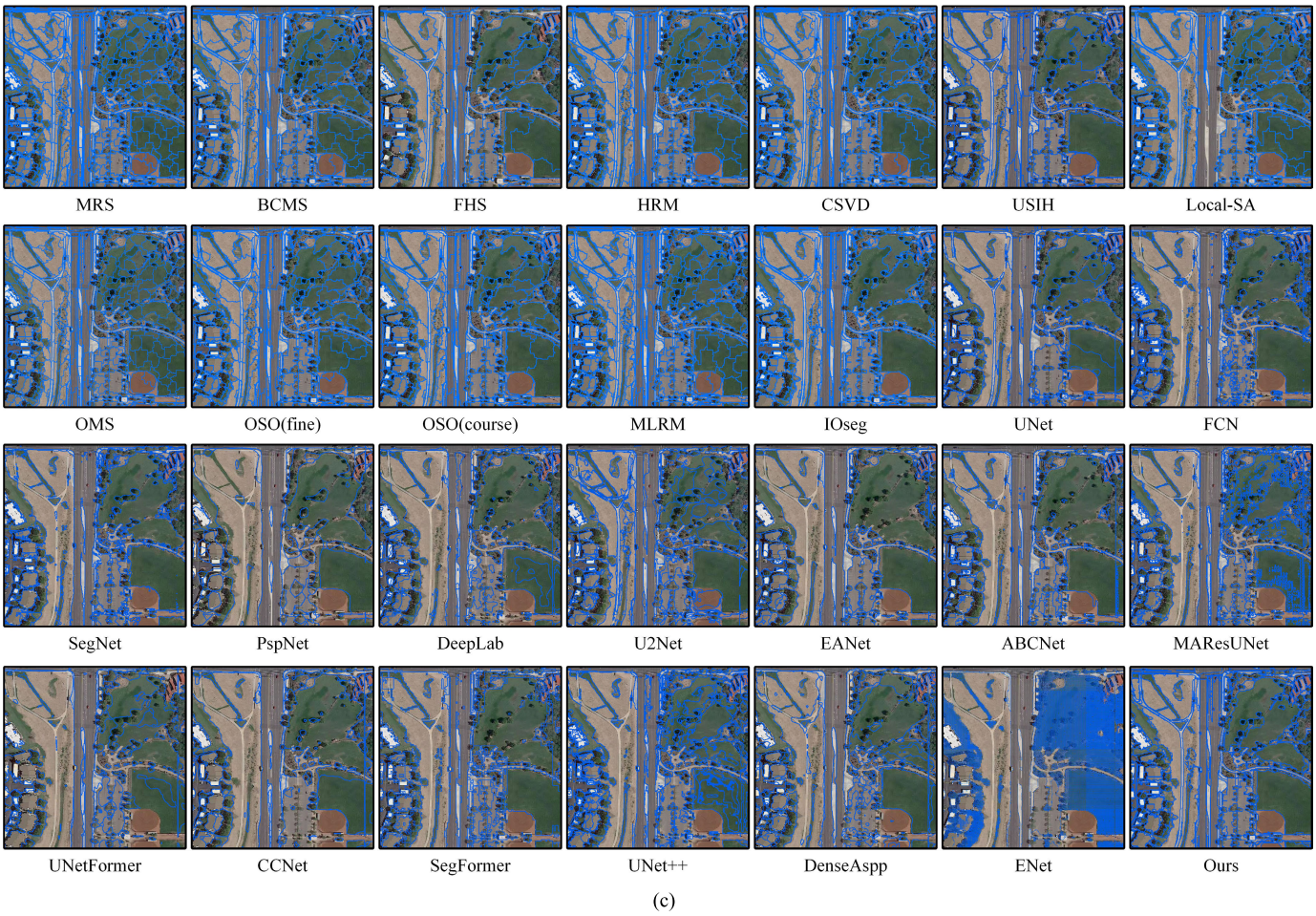

Fig. 14. *(Continued.)* (c) Image segmentation results in rural green spaces.

significant variations in size, texture, shape, and spectrum. DeepMerge successfully delineates large areas of bare land, factory buildings, roads, and even individual trees. Local-SA and USIH also present satisfactory segmented results, but some over-segmentation errors still exist (e.g., the fragmented areas within buildings in Local-SA and on bare lands in USIH). These qualitative analyses provide strong support for the robustness of DeepMerge.

V. Discussion

This article introduces DeepMerge, a novel deep-learning-based region merging method for remote sensing image segmentation. While our method utilizes only 0.19% of the total superpixels for training, it achieves compelling image segmentation results across various scenarios. Furthermore, the optimal scale parameter consistently stabilizes at 0.5 across the study area, eliminating the need for user-defined parameter tuning. This inherent advantage distinguishes DeepMerge from other multiscale segmentation methods, which typically require careful scale parameter selection. A near-zero scale parameter indicates high similarity, whereas values approaching or exceeding one signify low similarity. From Fig. 13, we notice that segmentation results in the scale range of [0.4, 0.6] are desirable for many applications, and the proposed DeepMerge can generate segmentation results in different scale parameters as needed. The computational cost and the parameter size of the proposed DeepMerge model are 81.56 GFLOPs and 348.42 MB, respectively.

To further elaborate on the significance of these findings, we analyze the main contributions of this work in detail.

1) Existing region merging methods often struggle to capture the complex relationships between superpixels at different scales. This work represents the first integration of RAG with deep learning for remote sensing image segmentation. As demonstrated in our experiments, DeepMerge achieves high segmentation accuracy, successfully segmenting both small and large geo-objects, and outperforming SOTA methods. This highlights the power of combining RAG with deep learning for capturing intricate spatial context and achieving accurate segmentation results in complex remote sensing scenes.
2) Unlike existing methods, DeepMerge exhibits a remarkable property: the optimal scale parameter is interpretable and predictable, consistently stabilizing at 0.5 across diverse scenarios. This finding addresses a critical challenge the selection of optimal scale parameters in region merging algorithms. Traditional methods often necessitate manual tuning and experimentation

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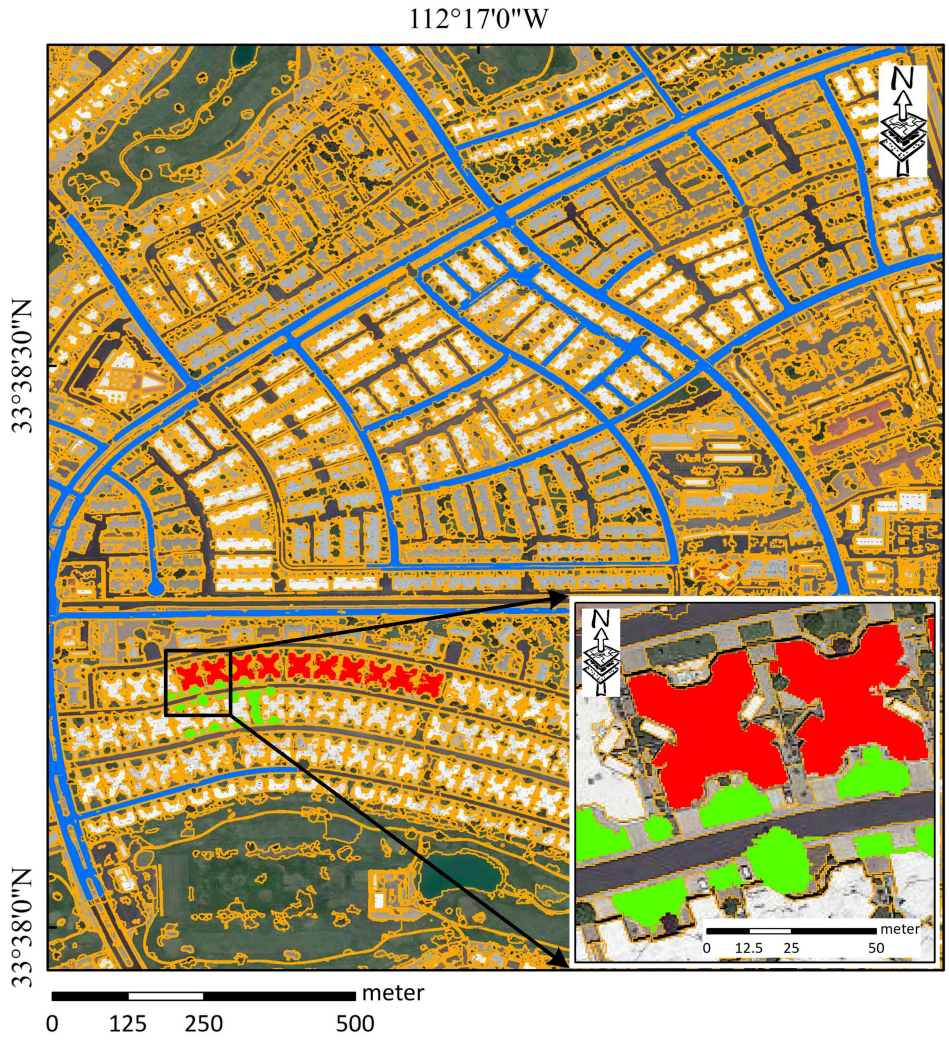

Fig. 15. Case of geo-objects with various sizes generated by DeepMerge for presentation.

to determine the appropriate scale, which can be time-consuming and subjective. DeepMerge, by demonstrating a stable optimal scale, eliminates this burden, offering a more objective and robust approach.

3) The efficiency of training data preparation is another key advantage of DeepMerge. Manual annotation of large-scale, high-resolution remote sensing images is a notoriously time-consuming and labor-intensive process. As highlighted in the introduction, annotating five cityscapes images can take 7.5 h. In stark contrast, DeepMerge's weakly supervised sample annotation method significantly accelerates this process. A proficient operator can annotate a training image (approximately 78 times larger than a cityscapes image) within the same 7.5-h timeframe. This represents a dramatic improvement in annotation efficiency, reducing the time and cost associated with training data preparation.

Many region-merging methods incorporate geo-object size as a parameter within their merging criteria to enhance processing efficiency. Although such a strategy improves the merging efficiency and yields even-sized geo-objects, it fails to meet the demand of many applications due to the uneven size distribution of geo-objects in an image. For instance, our dataset includes relatively small geo-objects, such as individual trees and narrow, intermittent sidewalks only three pixels wide, alongside substantially larger geo-objects like factory buildings and roads, which often comprise thousands of pixels. Fig. 15 shows an output example with small and big geo-objects in a large area. The blue mask in Fig. 15 is the road as a whole geo-object with 166 981 m^2. The polygons in red are independent residential houses with an average area of 1000 m^2. In the green areas, there are individual lawns, in which the smallest area of lawn only covers 42 m^2. This example highlights the substantial size difference, with the road area being 3976 times larger than the smallest lawn. In this study, the segmentation results of DeepMerge are able to capture the true sizes of geo-objects. This capability to accurately segment objects of vastly different sizes further underscores the effectiveness of DeepMerge in handling the complexities of real-world remote sensing data.

VI. Conclusion

This study presents DeepMerge, a novel deep-learning-based method for stepwise optimized image segmentation of

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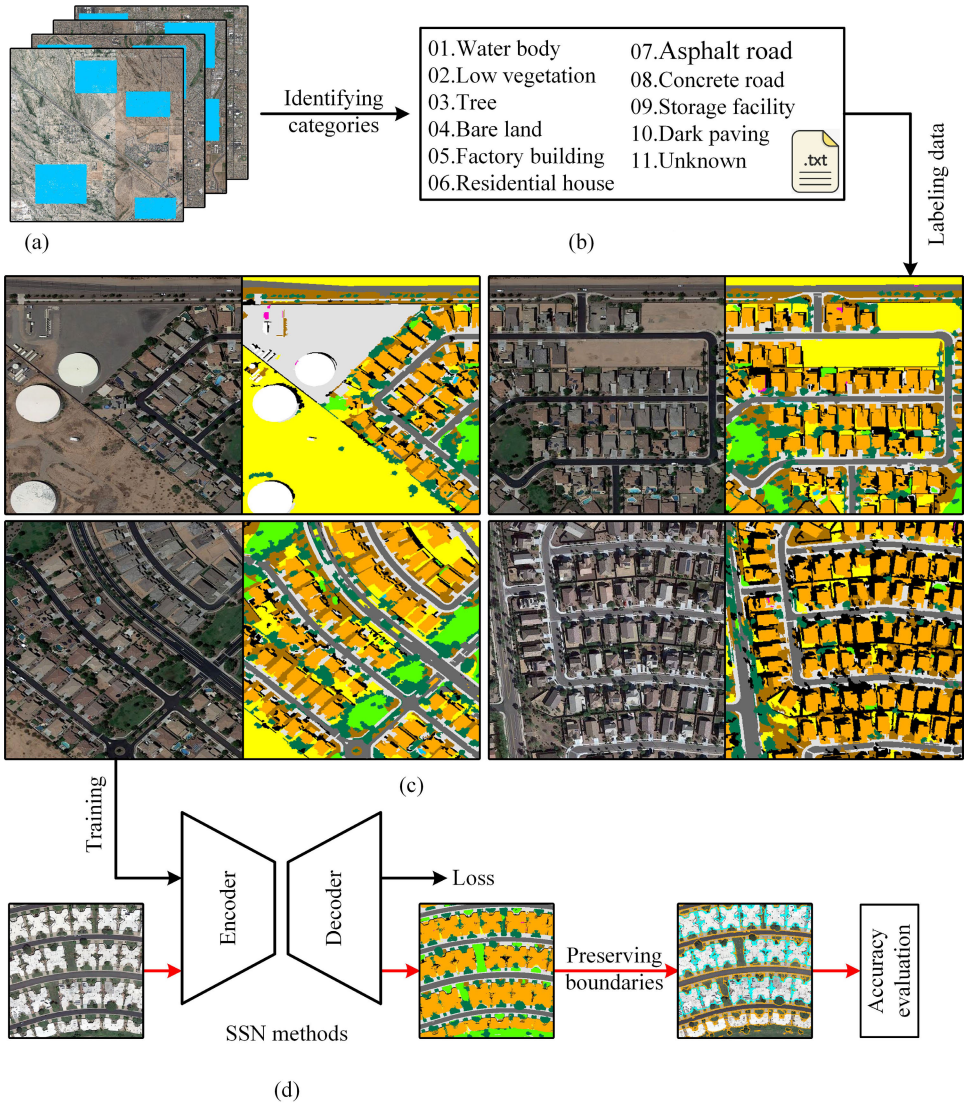

Fig. 16. Workflow about image segmentation based on SSN. (a) Training data area corresponding to Fig. 11, (b) land cover categories in the dataset, (c) training dataset for semantic segmentation, and (d) processing of remote sensing image segmentation based on SSN methods.

large-area, high-resolution remote sensing imagery. Given an initial super-pixel segmentation result, DeepMerge automatically obtains optimal segmentation output in a stable scale parameter.

We proposed the cross-scale embedding, cross-scale attention mechanism, and interpretable scale parameter to form CSFormer to capture the cross-scale information. Furthermore, a segment-based feature embedding module enhances the network's ability to preserve geo-object features. Evaluated in the Phoenix metropolitan area, DeepMerge demonstrates SOTA performance, qualitatively and quantitatively. The proposed method requires a small number of training samples with respect to the total number of super-pixels. In our experimental analysis, only 0.19% of the total number of super-pixels were labeled. DeepMerge based on the low ratio training dataset, achieves high advancements of 0.1 in F value, a large decrease of 0.04 in TE against SOTA methods. The optimal scale parameter of the proposed method stabilizes at 0.5. This makes the selection of the optimal parameter value easier for users. The proposed DeepMerge is suitable for the precise segmentation of large-area and VHR remote sensing images. It will provide efficient and precise segmentation of products. However, it still requires training samples, leading to time-consuming segmentation in small areas.

We plan to direct our further works to the potential improvement and the wide applications of the proposed DeepMerge by exploring the possibility of developing unsupervised DeepMerge-based segmentation approaches and further evaluating DeepMerge's performance on various land-cover classification problems.

APPENDIX

Fig. 16 illustrates the workflow for evaluating semantic segmentation neural networks (SSNs), encompassing dataset labeling, SSN training, boundary preservation, and accuracy assessment. To maintain consistency between the image segmentation and semantic segmentation training data, all superpixels within the blue tiles (see Fig. 11) were selected for SSN training. These regions were manually labeled into 11 categories [see Fig. 16(b)], resulting in a training dataset

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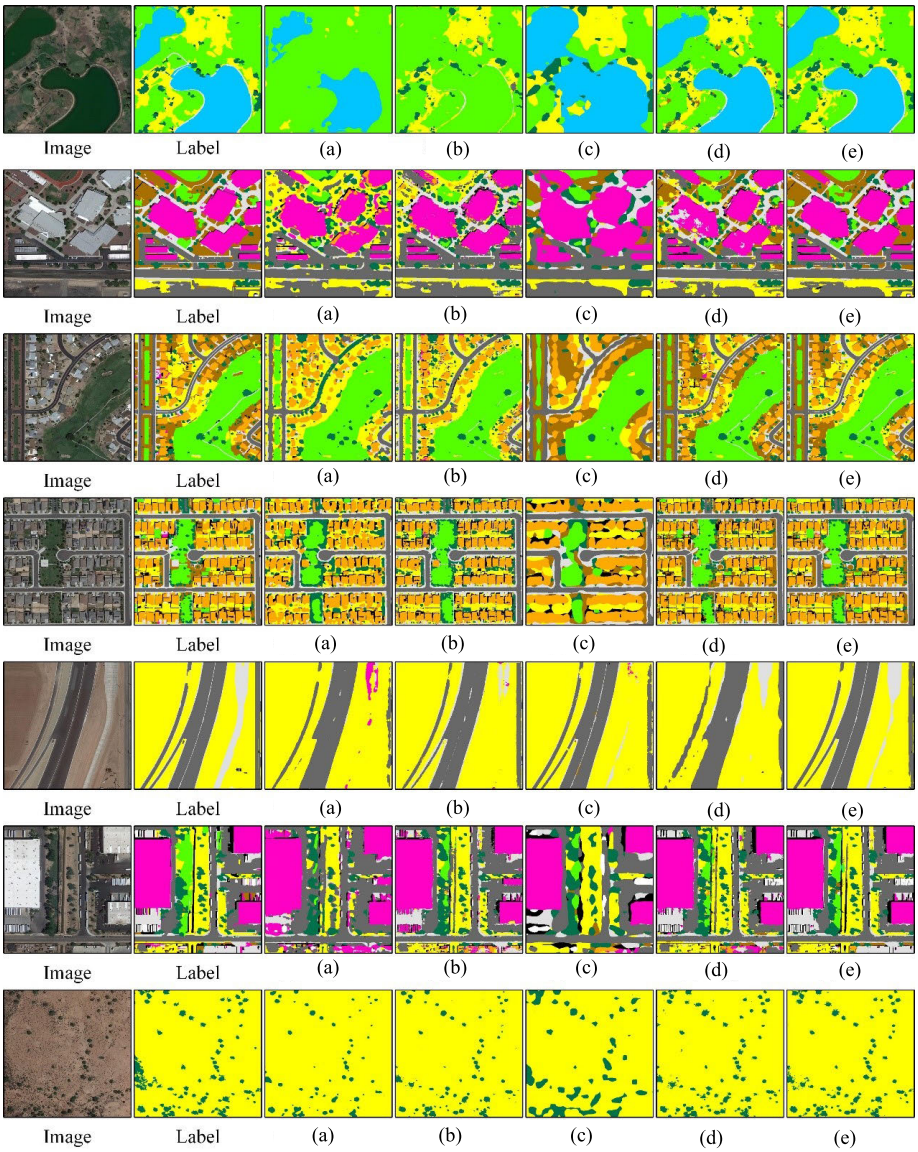

Fig. 17. Semantic segmentation results of five standard neural networks. (a) DeepLabV3. (b) FCN. (c) PSPNet. (d) SegNet. (e) UNet.

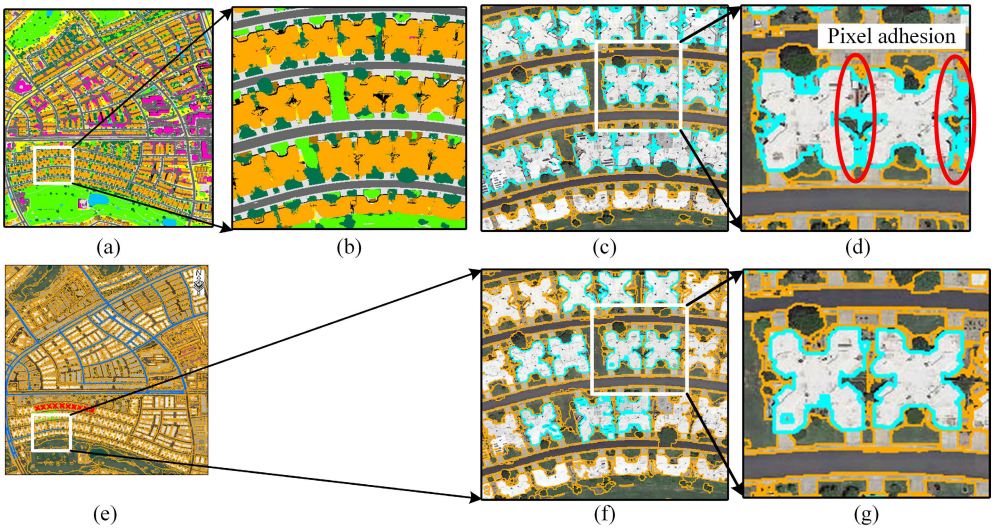

Fig. 18. Pixel adhesion of semantic segmentation methods. (a) and (e) Output of semantic segmentation and image segmentation, (b) zoomed-in view diagram with a local region, (c) and (f) image segmentation results, (d) image segmentation result of SSN with pixel adhesion, and (g) image segmentation of DeepMerge.

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of 1803 images, each with dimensions of 512 × 512 pixels. Upon completion of SSN training convergence, the output within the tested areas (green tiles in Fig. 11) represents the pixel-wise category prediction. To facilitate comparison with other segmentation methods, only the predicted boundaries from the SSNs were preserved and converted into standard vector-based segmentation outputs [see Fig. 16(d)].

While semantic segmentation results on the dataset appear promising based on pixel-wise the F-score within the test area (see Fig. 17), with UNet, for example, achieving a high F-score of 0.7085, image segmentation evaluation relies on polygon-based metrics. Thus, F value, as employed here, is calculated using object-based matching rather than pixel-based comparisons. Precision and recall measures are calculated using region overlapping. The precision measure's matching direction is defined as a reference-to-segment directional correspondence. Conversely, for the recall measure, we reverse this process by matching segments to reference geo-objects.

As illustrated in Fig. 18, pixel adhesion frequently occurs between similar, neighboring geo-objects in semantic segmentation results. Although the semantic segmentation results in Fig. 18(b) appear visually and quantitatively promising, deriving image segmentations solely from their boundaries leads to inferior results, contributing to the lower F values observed in Table IV.

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